

The core difference is that a `target` of **ES2017** includes support for **async/await** functions natively in the compiled JavaScript output, whereas **ES6** (ES2015) transpiles them into more complex generator-based code using Promises. [GitHub +1](#)

The `target` compiler option in `tsconfig.json` determines the ECMAScript version of the emitted JavaScript code and the default set of built-in API type definitions (`lib`) available during compilation. [TypeScript +1](#)

Feature Differences in Emitted Code

Feature 🔗	<code>target: ES6 (ES2015)</code>	<code>target: ES2017</code>
Arrow functions (<code>() => {}</code>)	Left intact (supported in ES6)	Left intact
async/await	Transpiled into generator functions and Promises, resulting in more verbose output	Left intact (supported in ES2017), resulting in cleaner, more readable code closer to the source
<code>Object.getOwnPropertyDescriptors()</code>	Not supported natively in compiled output (might require a shim/polyfill)	Supported natively in compiled output
<code>String.prototype.padStart</code> / <code>padEnd</code>	Not supported natively (might require a shim/polyfill)	Supported natively

`lib` Option Default

Setting the `target` option also sets the default `lib` compiler option value. [TypeScript](#)

- `target: ES6` defaults to including type definitions for ES6 features (like `Map`, `Set`, `Promise`, etc.).
- `target: ES2017` defaults to including all ES6 and ES2017 type definitions, providing type awareness for newer APIs like shared memory and specific string methods. [TypeScript +3](#)

Summary of Choice

- Choose **ES6** if your code needs to run in environments that support ES6 features but not ES2017 features (though most modern environments now support ES2017+).
- Choose **ES2017** (or higher) to take advantage of modern JavaScript features like **async/await** natively, resulting in smaller and cleaner output files and leveraging better

browser support for modern standards. You can check feature compatibility for specific environments on the Kangax compatibility table